

EMMANUEL NWORIE

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Professional Summary

Haberman Fellow and Quantitative Researcher pursuing a PhD in Computational and Applied Mathematics at Southern Methodist University with expertise in quantitative modeling, scientific computing, statistical analysis, and machine learning. Experienced in developing predictive mathematical models, large-scale computational simulations, and analytical frameworks for complex systems. Proven ability to integrate mathematical theory, data analysis, and computational methods to generate actionable insights across research, analytics, and quantitative decision-making environments.

Core Competencies

Quantitative Modeling | Scientific Computing | Predictive Analytics | Statistical Modeling | Machine Learning | Numerical Simulation | Data Analysis | Optimization | Python | MATLAB | R

Technical Skills

Programming: Python, MATLAB, R, SQL

Machine Learning & AI: Machine Learning, Deep Learning, Neural Networks, Classification, Regression, Feature Engineering, Model Evaluation, Predictive Analytics, Artificial Intelligence

Statistical Methods: Statistical Modeling, Statistical Inference, Time Series Analysis, Probability, Data Analysis

Scientific Computing: Partial Differential Equations, Finite Element Methods, Numerical Simulation, Dynamical Systems, Multilevel Methods, Optimization, Parameter Estimation

Libraries & Tools: NumPy, Pandas, SciPy, Scikit-Learn, TensorFlow/Keras, Matplotlib, Git, Linux, Jupyter, LaTeX

Research Experience

Doctoral Researcher

2021–Present

Southern Methodist University

- Develop predictive mathematical models and computational frameworks to study complex transport, growth, and competition phenomena in biological systems.
- Derive and analyze nonlinear partial differential equation models governing dynamic processes, system evolution, and emergent behavior.
- Design and implement finite-element simulation frameworks in Python to investigate large-scale transport mechanisms and evaluate competing scientific hypotheses.
- Integrate experimental datasets with mathematical models to perform parameter estimation, model calibration, uncertainty assessment, and quantitative validation.
- Apply statistical and computational techniques to analyze model performance and extract insights from high-dimensional biological data.
- Collaborate with interdisciplinary research teams at Southern Methodist University and Memorial Sloan Kettering Cancer Center to develop data-driven quantitative models.
- Present original research findings at national conferences, including meetings of the American Physical Society.

Selected Projects

Diffusion-Limited Growth Modeling and Simulation

- Developed finite-element simulation frameworks for diffusion-driven transport and growth processes using Python and numerical methods.
- Investigated transport-limited dynamics, competition mechanisms, and saturation behavior through computational experiments and mathematical analysis.
- Compared model predictions with experimental observations to evaluate competing biological hypotheses and improve model fidelity.

Customer Churn Prediction

- Developed and evaluated machine learning models, including neural network architectures, to predict customer attrition using structured business datasets.
- Performed feature engineering, data preprocessing, exploratory analysis, and model comparison across statistical and machine learning approaches.
- Assessed model performance using classification metrics and identified key factors influencing customer retention and churn behavior.
- Translated predictive modeling results into actionable insights for customer retention and business decision-making.

Heart Disease Prediction

- Developed predictive classification models using healthcare datasets and statistical learning techniques.
- Performed feature selection, data preprocessing, model training, and validation to improve predictive performance.
- Analyzed feature importance and model outputs to identify variables associated with elevated cardiovascular risk.

Artificial Intelligence and Deep Learning

- Developed neural network and deep learning models for supervised learning and predictive analytics applications.
- Applied TensorFlow/Keras frameworks for model development, optimization, and performance evaluation.
- Investigated feature extraction, classification, and model generalization techniques using real-world datasets.
- Applied concepts from artificial intelligence, machine learning, and statistical learning to solve practical analytical problems.

Selected Honors

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| • Haberman Fellowship, Southern Methodist University | 2022–2027 |
| • Gold Medalist, National Mathematics Competition for University Students (NAMCUS), Nigeria | 2018 |
| • Outstanding Graduating Student, Department of Mathematics, University of Nigeria, Nsukka | 2018 |
| • Outstanding Graduating Student, Faculty of Physical Sciences, University of Nigeria, Nsukka | 2018 |

Awards and Recognition

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| • Betty McKnight Graduate Teaching Award | 2023–2024 |
| • Employee of the Year, Campus Recreation, Southern Methodist University | 2025–2026 |
| • Multiple Conference Travel Grants for Research Presentations | |

Teaching & Leadership

Graduate Tutor

2024–Present

Altshuler Learning Enhancement Center (ALEC), Southern Methodist University

- Provide individualized academic support across the undergraduate mathematics curriculum, including calculus, linear algebra, differential equations, probability, statistics, and other quantitative courses.
- Assist students from diverse academic disciplines in developing mathematical reasoning, problem-solving strategies, and quantitative literacy.
- Support student success through one-on-one tutoring, conceptual instruction, and examination preparation.
- Help students strengthen analytical thinking and quantitative skills applicable across STEM, business, economics, and social science programs.

Graduate Teaching Assistant, Southern Methodist University

2021–2024

- Supported instruction in mathematics and statistics courses and developed Python/MATLAB computational demonstrations.

Head of Mathematics Department, Doyen Academy and Doyen British High School

Jan 2019–Jul 2021

- Led the Mathematics Department and coordinated curriculum delivery and instructional activities.
- Taught Cambridge Advanced Level Mathematics, Statistics, and Mechanics.
- Prepared students for Cambridge A-Level examinations, with multiple students achieving distinction-level performance.

Education

Southern Methodist University

PhD, Computational & Applied Mathematics

Expected 2027

M.S., Computational & Applied Mathematics

2023

University of Nigeria, Nsukka

B.S. Mathematics (Honors)